the body frame intrinsic rotations are intuitive to me. forget the quaternions for the moment, and let's go deeper into how intrinsic ZYX is equal to Extrinsic XYZ

Intrinsic Z-Y'-X" = Extrinsic X-Y-Z: The Deep Proof

Let's build this from scratch, carefully.

Setup

We want to apply three rotations with angles α, β, γ around Z, Y, X in that order — **intrinsically** (body frame).

Call the rotation matrices:

- **A** = Rz(α)
- **B** = Ry(β)
- **C** = Rx(γ)

Step 1: First Rotation (Intrinsic Z)

No ambiguity here — body frame and world frame are aligned at the start.

```
R₁ = A = Rz(α)
```

The body frame has now rotated. Its new axes are the columns of A.

Step 2: Second Rotation (Intrinsic Y')

Now the body's Y axis **is no longer the world Y**. It has been rotated by A.

The body's Y axis in world coordinates is: $\mathbf{A} \cdot \hat{\mathbf{y}}$

You want to rotate by angle β around this tilted axis.

Using the conjugation formula — to rotate around a rotated axis, you:

1. Undo A (go back to world frame)
2. Apply Ry(β) around the simple world Y

3. Redo A

```
R₂ = A · B · A⁻¹
```

So after two intrinsic rotations, the total rotation is:

```
R₁₂ = R₂ · R₁ = (A · B · A⁻¹) · A = A · B
```

The $A^{-1} \cdot A$ cancels perfectly. ✓

Step 3: Third Rotation (Intrinsic X")

Now the body's X axis has been rotated by **both** A and B.

The body's X axis in world coordinates is: $\mathbf{A \cdot B \cdot \hat{x}}$

To rotate by γ around this doubly-rotated axis, conjugate with the combined rotation so far (A·B):

```
R₃ = (A·B) · C · (A·B)⁻¹
```

Total rotation after all three intrinsic steps:

```
R_total = R₃ · R₁₂
        = [(A·B) · C · (A·B)⁻¹] · (A·B)
        = (A·B) · C · [(A·B)⁻¹ · (A·B)]
        = A · B · C
```

Again, the inverse cancels. ✓

The Result

```
R_intrinsic(Z→Y'→X'') = A · B · C = Rz(α) · Ry(β) · Rx(γ)
```

Now: What is Extrinsic X-Y-Z?

Extrinsic means **all axes are fixed world axes**, applied in stated order: first X, then Y, then Z.

When composing extrinsic rotations, each new rotation **pre-multiplies** (goes on the left), because it acts on the result of everything done so far in the fixed frame:

```
R_extrinsic(X→Y→Z) = Rz(α) · Ry(β) · Rx(γ)
```

- Apply Rx(γ) first (rightmost)
- Then Ry(β)
- Then Rz(α) (leftmost)

They Are Identical

```
R_intrinsic(Z→Y'→X'') = Rz · Ry · Rx = R_extrinsic(X→Y→Z)
```

The **same matrix**, arrived at by two completely different physical stories.

Why the Cancellation Always Works

The pattern is not a coincidence — it's structural. Every intrinsic step follows this template:

> *"To rotate around an axis that has been moved by prior rotations M, conjugate: $M \cdot R_n \cdot M^{-1}$, then compose with M"*

```
(M · Rₙ · M⁻¹) · M = M · Rₙ
```

The $M^{-1} \cdot M$ **always** cancels. This means each intrinsic rotation simply **appends** its matrix to the right of the running product — regardless of how complex M has become.

Extrinsic rotations do the opposite: each new rotation appends to the **left**. So reversing the stated order makes them identical.

Side-by-side Summary

```
Intrinsic Z→Y'→X''      Extrinsic X→Y→Z
────────────────────     ────────────────────

Start:    A              Apply Rx(γ):    Rx
After Z:  A              Apply Ry(β):  Ry·Rx
After Y': A·B            Apply Rz(α):Rz·Ry·Rx
After X'':A·B·C

Both = Rz(α)·Ry(β)·Rx(γ)  ✓
```

The intrinsic story builds **right to left in application, left to right in writing**.
The extrinsic story builds **left to right in application, right to left in writing**.
They meet in the middle at the same matrix.

can you explain for Step 2 where you get the sequence to rotate around a rotated axis

Great question — this is the heart of it. Let me derive the conjugation formula from scratch.

The Problem

After the first rotation A = Rz(α), the body's Y axis is no longer the world Y. Call this new tilted axis **û**.

You want a rotation matrix that rotates by angle β around **û** (expressed in world coordinates).

Key Insight: You Already Know How to Rotate Around World Axes

You know Ry(β) rotates by β around the **world Y axis** ŷ.

The question is: **can you re-use Ry(β) to rotate around û instead?**

Yes — by changing coordinates.

The Coordinate Change Trick

Here's the core idea:

> If you can temporarily **transform û back into ŷ**, do the rotation there, then **transform back** — you've effectively rotated around û.

In three steps:

1. Rotate so that û aligns with world Y

The transformation that moved ŷ → û was A. So A^{-1} undoes it:

```
A⁻¹ · û = ŷ
```

Apply A^{-1} to the entire world — now û is sitting where ŷ used to be.

2. Apply Ry(β) in this simplified world

Now that û coincides with ŷ, rotating around ŷ is the same as rotating around û:

```
Ry(β)
```

3. Undo the coordinate change

Reapply A to bring everything back to the original world orientation:

```
A
```

Composing the Three Steps

A vector **v** in the world frame gets transformed as:

```
v' = A · Ry(β) · A⁻¹ · v
```

So the matrix that represents "rotate by β around the axis û = A·ŷ" is:

```
R_around_û = A · Ry(β) · A⁻¹
```

This is the **conjugation formula.** It works for any rotation matrix and any axis.